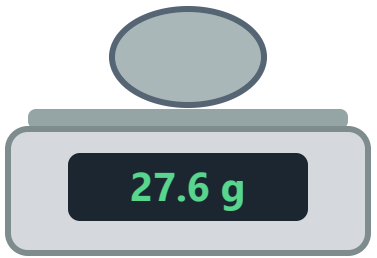
Measuring the density of a solid (water displacement)

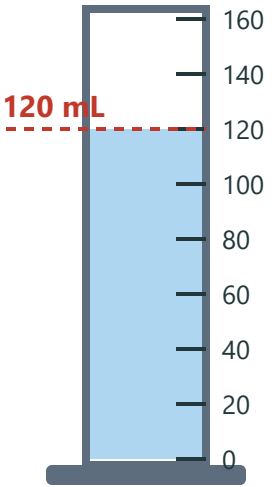
1



Weigh the solid

m = 27.6 g

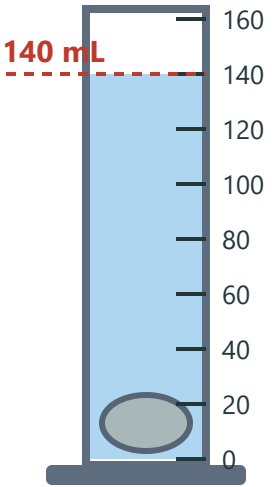
2



Water in a graduated cylinder

V_1 = 120 mL

3



Drop the solid in: the water rises

V_2 = 140 mL

4

$V = V_2 - V_1 = 140 - 120 =$ **20 mL**

$$d = \frac{m}{V} = \frac{27.6\ g}{20\ mL}$$

d = 1.38 g/mL